

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B.TECH, Branch : Mechanical Engineering/Mechanical Engineering (Sandwich) **Semester : VI**

Subject Code & Name: BTMEC602_Y19 , Machine Design-II

Max Marks: 60

Date: 02/07/25

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)	1/1	12
1	Bearings are used to ----- a.) Reduce the friction b. Transmit the load c. Support the shaft d. All of the above		1
2	Taper Roller Bearings are used to ----- a. Resists the Axial force b. Transmit the radial load c. Transmit the axial load d. Nullify the Axial forces		1
3	Two parallel shafts can be connected by means of ----- a. Spur Gears b. Helical Gears c. Bevel Gears d. a & b.		1
4	The drives that transmits power by means of friction are --- a. Gear drives b. Belt drive & Chain drive c. Rope & chain drives d. Rope & Belt drives		1
5	In a pair of bevel gears, in the “Çrown Gear” arrangement the pitch angle is -----degree, a. 90 b. 45 c. 60 d. 30		1
6	Worm gears are generally used for connecting two shafts at---- a. right angle b. inclined c. parallel d. perpendicular but non intersecting		1
7	In case of centrifugal clutches the centrifugal force acts in ----- direction a. Radial inward b. Tangential c. Radial outward d. None of the above		1
8	In case of heavy vehicle trucks the clutch used is ----- a. Single plate b. Multi-plate c. Cone clutch d. Centrifugal clutch		1
9	In automobile vehicles the brakes generally used to ----- a. Retard the motion b. Stop the vehicle c. Neither a nor b d. Either a or b		1

10	When brakes are applied on a moving vehicle; the kinetic energy is converted to-----					1
	a. Heat energy	b. Mechanical energy	c. Potential energy	d. Electrical energy		
11	The effectiveness of the brakes depends on -----					1
	a. Area of brake linings	b. Radius of car wheel	c. Amount of pressure applied to shoe brakes	d All of the above		
12	Hand brake is applicable to-----					1
	a. only rear wheels	b. only front wheels	c. both front and rear wheels	d. all of the above		
Q. 2	Write a short notes on :-				1/2	12
A)	Viscosity & Viscosity Index ,					6
B)	Types and construction of Rolling contact Bearing.					6
Q.3	Solve the following.					12
A)	A single row deep groove ball bearing has a dynamic load carrying capacity of 50,000 N and operates on following work cycle:- Radial load of 5000 N at 500 rpm for 25% of time. Radial load of 8000 N at 600 rpm for 50% of time and, Radial load of 4000 N at 450 rpm for remaining 25 % of time . Calculate the expected life of bearing in hours.				2/3	6
B)	A journal bearing 0.080 m long and 0.05 m diameter running at 1000 rpm is subjected to a radial load of 5000 N. The bearing temperature is 85 ⁰ C and ambient temperature is 30 ⁰ C. The oil used is having viscosity of 0.0128 Kg/m-s at operating temp of 30 ⁰ C. The bearing is air cooled with diametric clearance of 0.05 mm. Calculate the coefficient of friction considering leakage clearance of 0.003.				2/3	6
Q. 4	Solve Any Two of the following.					12
A)	Explain with help of neat diagram classification & selection criteria of type of Gears.				1/2	6
B)	A pair of spur gears with 20 degree full depth involute teeth consists of pinion having 20 teeth meshing with a gear having 38 teeth. The module is 3 and face width is 34 mm. The material of pinion and gear is steel with a ultimate tensile strength of 400 N/mm ² . The gears are heat treated to surface hardness of 300 BHN .The pinion rotates at 1400 rpm and service factor for the application is 1.25. Assume the velocity factor accounts for the dynamic load and factor of safety is 1.5 . Determine the power that the gear can transmit. Assume Lewis form factor Y =0.33.				2/3	6

C)	A pair of bevel gears having 20° pressure angle consists of a pinion having 20 teeth meshing with a gear having 36 teeth. The face width is 20 mm and module is 4. The material of gear and pinion is steel 50 C4 ($S_{ut} = 750 \text{ N/mm}^2$) The gear teeth hardness is 450 BHN grounded and lapped . The pinion rotates at 500 rpm and receives 4 Kw power from electric motor. The starting torque is 150 % of the rated torque. Find out the effective load acting on the gears. Use following values if required $c = 11400 \text{ N/mm}^2$, $e = 0.0125$	2/3	6
Q.5	Solve Any Two of the following.		12
A)	With the help of neat diagram Classify the Bevel gears based on pitch angle.	1/2	6
B)	A pair of straight bevel gears mounted on shaft are intersecting at right angle. The number of teeth on pinion and gear are 18 and 24 respectively. The pressure angle is 20 degree. The pinion shaft is connected to an electric motor developing 4 Kw. power at 1400 rpm . The service factor is 1.25 . The pinion and gears are made of steel having ultimate strength of 600 N/mm^2 and heat treated to surface hardness of 400 BHN. The manufacturing error between meshing teeth is 10 micro meter. The module and face width are 3 mm and 15 mm respectively. Find out the factor of safety against bending as well as pitting failure. Assume form factor $Y = 0.345$ and $C = 11400 \text{ N/mm}^2$	3/5	6
C)	A centrifugal clutch transmits 18 Kw at 700 rpm consists of 4 shoes. The clutch is to be engaged at 500 rpm. The inner radius of drum is 170 mm. The center of gravity of shoe is 140 mm when it is engaged. The coefficient of friction is 0.25, while the intensity of pressure is 0.15 N/mm^2 , Calculate 1) mass of each shoe, 2) Dimensions of friction lining	2/3	6
Q. 6	Solve Any Two of the following.		12
A)	1)) Define the terms Belt Slip & Creep. 2) Differentiate between Flat belt and V-belt drives.	1/2	6 (3X2)
B)	Design a flat belt drive to operate dynamo generating 18 Kw at 1800 rpm. The pulley mounted on it is having diameter of 200 mm. Use the following data:- 1) Allowable stress for the belt $= 2.5 \text{ Mpa}$. 2) Density of rubber belt $= 1000 \text{ Kg/m}^3$ 3) Thickness of the belt $= 10 \text{ mm}$ 4) Angle of contact at dynamo pulley $= 160^\circ$ 5) Coefficient of friction between belt and pulley $= 0.25$ 6) Dynamo efficiency $= 80\%$	2/3	6
C)	A worm gear drive 20 Kw power running at 1800 rpm is used to operate a machine carriage at 100 rpm . The worm is triple start and 60 mm pitch diameter, whereas the worm gear has 80 teeth of 6 mm module. The tooth is 20 degree full depth involute & coefficient of friction between mating teeth is 0.1. Determine :- 1) Tangential force acting on the worm. 2) Axial thrust and separating force on worm and 3) Efficiency of the drive.	3/4	6
	*** End ***		

